

An audit of the immrep25 unseen-peptide benchmark

Signal-to-noise of CDR3 homology and V/J chain pairing across the positive set

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Abstract

We ask whether the *positive* TCR-peptide assignments in the immrep25 unseen-peptide benchmark carry epitope-specific signal, or only noise and publicity. Two biologically-grounded probes—within- vs. between-epitope CDR3 *homology* and per-epitope V/J *chain pairing*—are evaluated one-vs-many per epitope and aggregated across epitopes, then calibrated against cohorts of known quality (TCRvdb experimentally verified / refuted; VDJdb high- / low-reference-support) and three controls that fix the noise floor at S/N=1: two OLGA controls (raw random and generation-probability-matched) and a real post-thymic-selection bulk repertoire (AIRR). On both probes and both chains the immrep25 positives sit just above that floor and one to three orders of magnitude below every quality cohort; their weak homology is fully explained by public TCRs (it collapses to the floor once public TCRs are removed) and their inter-chain pairing shows essentially no coupling. We conclude the positive labels carry no meaningful epitope-specific signal beyond noise and publicity.

1 Data and design

Cohorts. Every cohort is reduced to a common per-chain schema (one CDR3 with its V/J and epitope) and a paired schema (CDR3_α, V_α, J_α, CDR3_β, V_β, J_β, epitope). The six cohorts span the expected signal-to-noise (S/N) range:

cohort	definition	role
TCRvdb true	TCRvdb, $p_{\text{adj}} < 10^{-5}$	experimentally verified (top)
VDJdb HQ	VDJdb human MHC-I, ≥ 2 references per (TCR, epitope)	strongly supported
VDJdb LQ	VDJdb human MHC-I, single reference	weakly supported
TCRvdb false	TCRvdb, $p_{\text{adj}} \geq 10^{-5}$	experimentally refuted
immrep25 pos	immrep25 <code>label=1</code>	system under test
AIRR random	real repertoire (isalgo/airr_control), random labels	real-data floor
AIRR rank-ladder	top-1000 real clonotypes; epitope = frequency-rank bin of 50	structured real control
OLGA pgen-ladder	50 per epitope, P_{gen} -matched to that immrep epitope	structured null
OLGA random	raw OLGA generation, random labels	noise floor

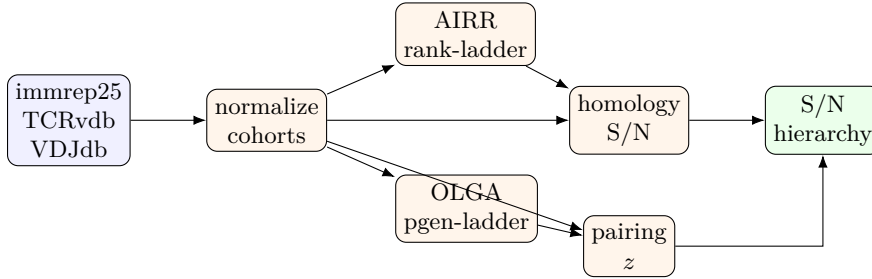
Cohorts draw on VDJdb [1], the functionally-validated TCRvdb/MATCHMAKERS database [2], the immrep25 unseen-peptide benchmark [3], and an OLGA control built from the TCR generation model of Murugan et al. [4]. The immrep25 test set covers 20 nine-mer peptides that are *absent from VDJdb*, with 50 positives (and 450 negatives) each; only positives are audited. VDJdb HQ is defined per chain (TRA and TRB separately) as a (TCR, epitope) supported by ≥ 2 distinct references, with a paired complex counted HQ if *either* chain qualifies. TCRvdb true/false are dominated by the same two epitopes (YLQPRTFLL, GLCTLVAML), giving an epitope-matched verified-vs-refuted contrast.

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One-vs-many. Every metric is computed *per epitope* against all others (each epitope with ≥ 30 records vs. the rest of its cohort) and then aggregated to a dataset value—geometric mean $\pm 95\%$ CI for homology S/N, mean $\pm 95\%$ CI for pairing—across epitopes. The signal therefore attaches to individual epitopes and the reported error bars are the epitope-to-epitope spread (n = number of qualifying epitopes); no bootstrap resampling is used.

Controls with realistic epitope structure. Random synthetic labels give S/N=1 by construction, so we also build controls whose *epitope grouping is structured*, to check whether any generic property could manufacture within-epitope homology. **OLGA pgen-ladder:** for each of the 20 immrep25 epitopes we draw 50 OLGA sequences whose per-chain P_{gen} matches *that* epitope’s positives (so each control epitope inherits the real epitope’s generation degree). **AIRR rank-ladder:** the top 1000 real clonotypes by clone frequency, cut into 20 consecutive rank bins of 50 that serve as epitopes (so each “epitope” is a set of equally-public clones). Two random-label floors (OLGA random, AIRR random) are kept for reference.

Pipeline.



2 Homology test

Metric. For a single chain we score each epitope one-vs-many. Only equal-length CDR3s can share a Hamming distance, so pairs are counted within length bins. For epitope e (N_e TCRs) against the other $N - N_e$ TCRs of its cohort,

$$P_{\text{self}} = \binom{N_e}{2}, \quad P_{\text{non}} = N_e (N - N_e)$$

(equal-length pairs); with $m_{\leq d}/l_{\leq d}$ the within-/against-rest pairs at Hamming $\leq d$ and an exposure-proportional pseudocount a (so S/N $\rightarrow 1$ under no enrichment),

$$S/N_e(d) = \frac{(m_{\leq d} + a)/P_{\text{self}}}{(l_{\leq d} + a P_{\text{non}}/P_{\text{self}})/P_{\text{non}}}.$$

The dataset value is the geometric mean of S/N_e over epitopes ($\pm 95\%$ CI); the error bars are thus the epitope-to-epitope spread. $d=0$ isolates publicity (identical CDR3s); $d=1$ is the substitution-neighbour convergence signal exploited by curated specificity databases [1]; signal $\Rightarrow S/N \gg 1$, noise $\Rightarrow S/N \approx 1$.

Result. Figure 1 orders the cohorts: VDJdb-HQ $>$ TCRvdb-true $>$ TCRvdb-false $>$ VDJdb-LQ $\gg$ immrep25 $>$ OLGA. On TCR β the verified cohorts reach S/N = 1756 (VDJdb-HQ) and 496 (TCRvdb-true), whereas immrep25 positives give only 2.69 and *every* control sits at the floor—including the two structured ones: the OLGA pgen-ladder (0.946), whose epitopes match immrep25’s per-epitope generation degree, and the AIRR rank-ladder (0.860), whose epitopes are bins of equally-public real clonotypes. Neither reproduces immrep25’s within-epitope homology,

so its excess is not manufactured by generation degree, by clone frequency, or by the real thymic-selection process (AIRR random 1.00); it is specific to immrep25’s epitope→TCR assignments (i.e. publicity, §4). The same holds on TCR α (immrep25 3.88). immrep25’s CI excludes 1 but lies one to three orders of magnitude below every quality cohort. As the substitution radius grows (Fig. 2) the verified cohorts stay high while immrep25 and all controls remain near the floor across $d = 1, 2, 3$.

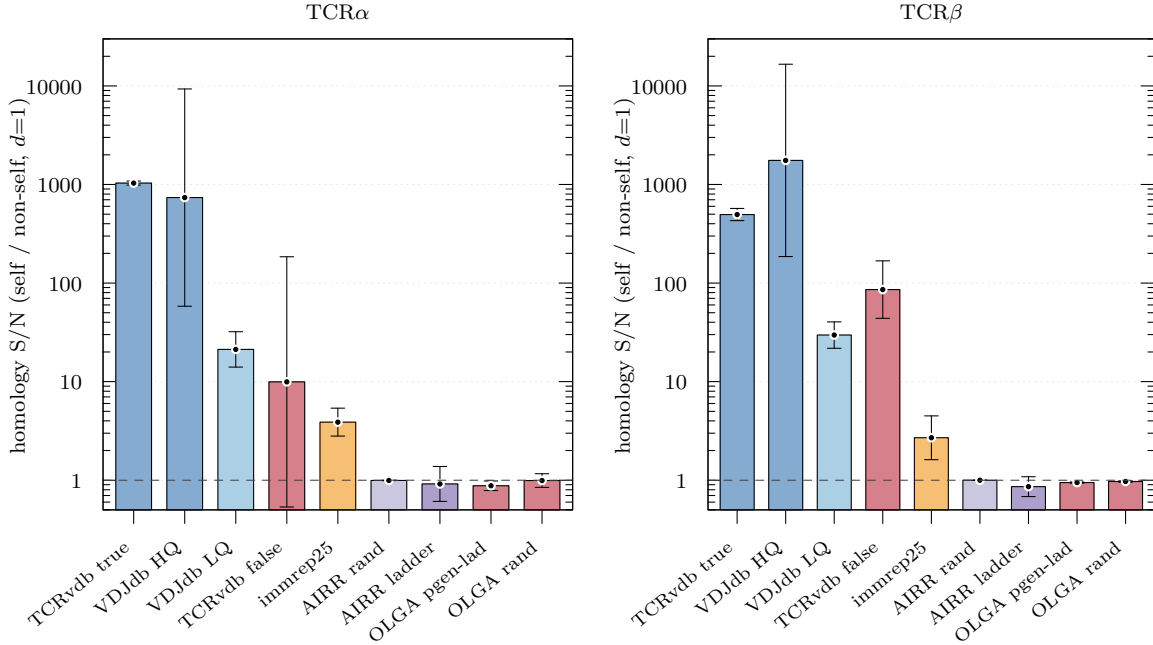


Figure 1: Homology S/N ($d=1$) per cohort, TCR α and TCR β (one-vs-many: each epitope with ≥ 30 records vs. the rest; bar = geometric mean over epitopes, whiskers = 95% CI = between-epitope spread). Dashed line: noise floor S/N=1.

By-epitope breakdown. Figure 3 plots the within-epitope neighbour rate for every epitope with $n \geq 30$ (one point per epitope; boxplots show the per-cohort quartiles); the vertical gap to each cohort’s non-self rate (red bar) is the S/N. immrep25’s points hug its non-self line, whereas the verified cohorts sit orders of magnitude above. Table 1 lists all 20 immrep25 epitopes: in total they contribute only 54 (TCR β) / 53 (TCR α) within-epitope Hamming-1 neighbours — versus 0 / 1 for the P_{gen} -matched OLGA control and tens of thousands for the VDJdb cohorts — and even that handful is concentrated in a few epitopes (e.g. YLFNADIWI) and is entirely public (§4). Tellingly, the AIRR rank-ladder carries *more* within-group α -neighbours (93) than immrep25 (53) yet its S/N stays below 1: those neighbours are public clones scattered across rank bins, so raw homology without epitope concentration does not lift the S/N.

Graph view. The S/N ratio is the within- vs. between-epitope edge-density of the homology graph (nodes=CDR3, edges=Hamming ≤ 1 ; all epitopes with ≥ 30 records). Figure 4 renders it (graphviz `sfdp` layout): the verified cohorts form dense epitope-coloured cliques, while the controls (even where they carry edges, e.g. the public AIRR rank-ladder) show no epitope-coloured clustering—their few edges cross colours rather than forming cliques.

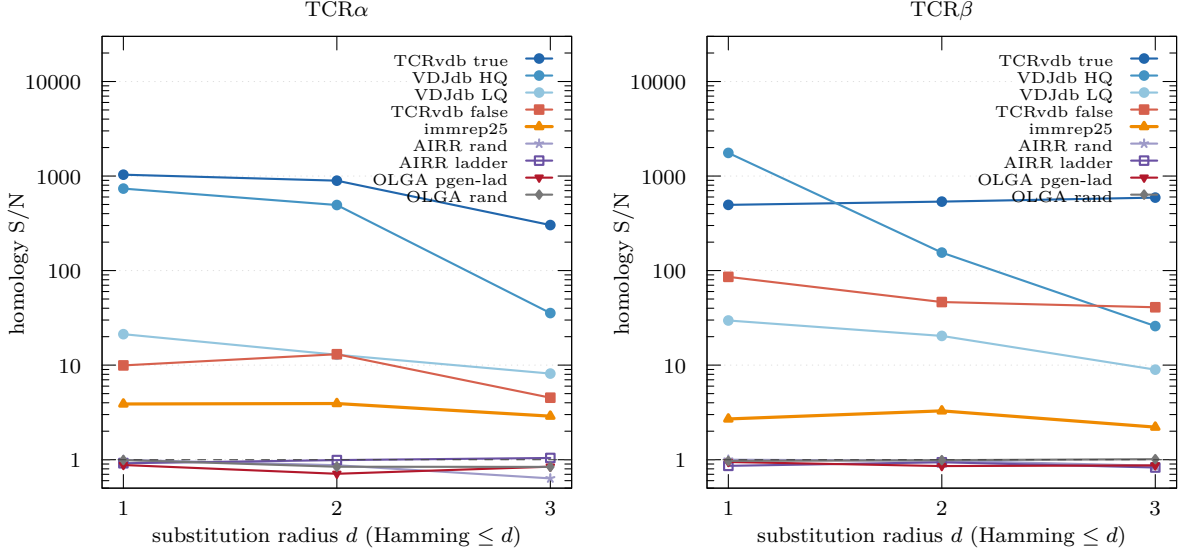


Figure 2: Homology S/N vs. substitution radius $d = 1, 2, 3$ (one-vs-many geometric mean). Verified cohorts stay high; immrep25 and all controls track the floor.

3 Pairing test

Metric. Genome-wide α/β pairing is close to random [5], but given an epitope the V/J usage may be constrained. For every epitope with ≥ 30 paired records we measure two signals as the *excess over a permutation null*, in bits, then average over epitopes ($\pm 95\%$ CI):

- **gene-usage bias:** $\overline{\text{KL}}(P_e \parallel Q)$ over axes $V\alpha, J\alpha, V\beta, J\beta$ (P_e = within-epitope gene distribution, Q = cohort background) minus its mean under size-matched random draws from Q .
- **inter-chain MI:** $I((V\alpha, J\alpha); (V\beta, J\beta))$ within epitope (Miller–Madow) minus its mean under $\alpha \leftrightarrow \beta$ shuffles (which keep the marginals but destroy coupling).

Both are 0 under no signal and grow (in bits) with genuine per-epitope structure. Writing X for either statistic and $\{X_{\text{null}}^{(k)}\}$ for its K permutation replicates, the excess is $\Delta X_e = X_e - \overline{X_{\text{null}}}$ and the standardized effect is

$$z_e = \frac{X_e - \overline{X_{\text{null}}}}{\text{sd}(X_{\text{null}})}.$$

We report the excess ΔX (in bits) rather than z , because for small epitopes the null variance can collapse ($\text{sd} \rightarrow 0$) and inflate z ; the excess is stable and keeps the natural unit.

Result. Figure 5 shows both. The verified cohorts carry substantial gene-usage bias (VDJdb-HQ 1.62 bits) and real inter-chain coupling (TCRvdb-true 1.04 bits). immrep25 positives are near the floor on both: gene-usage bias 0.203 bits and, decisively, an inter-chain coupling of only 0.047 bits—an order of magnitude below every quality cohort and barely above the AIRR and OLGA controls ($-0.006 / -0.001$ bits). The residual gene-usage bias is consistent with the public, HLA-restricted composition of the positives rather than epitope-specific pairing.

4 Publicity control

The one alternative to “pure noise” is *publicity*: immrep25 might reuse public TCRs whose chance homology mimics epitope signal. Although only 0.3% of positives share *both* chains exactly with

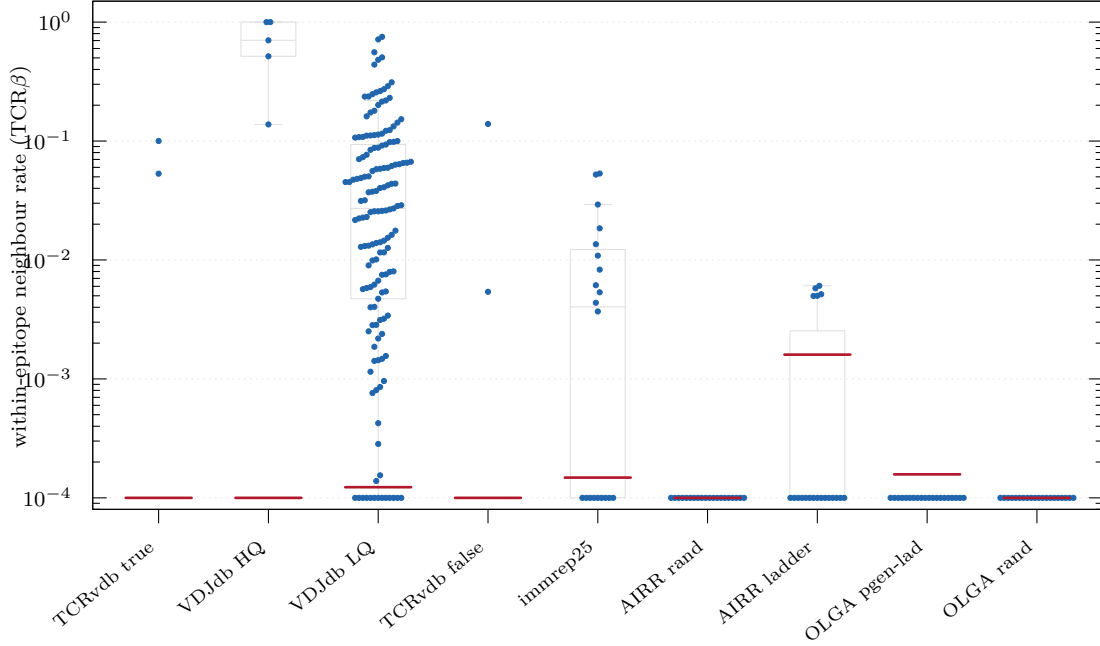


Figure 3: Per-epitope within-epitope neighbour rate ($\text{TCR}\beta$, $\text{Hamming} \leq 1$; each point is one epitope, $n \geq 30$; boxplots = per-cohort quartiles). Red bars = cohort pooled cross-epitope (non-self) rate; the vertical gap is the homology S/N.

VDJdb, allowing a single substitution and either chain, 74.5% of immrep25 positives are within $\text{Hamming} \leq 1$ of a known VDJdb/TCRvdb TCR (61.5% on α , 31.2% on β)—the α chain being especially germline-public. Consistently, the positives are enriched for high generation probability relative to raw OLGA sampling (Fig. 6), i.e. they are “easy-to-generate”, public-leaning sequences; this is exactly why the random control must be P_{gen} -matched.

Removing public positives collapses the (already weak) homology to the noise floor: one-vs-many S/N falls from 2.69 to 1.00 on $\text{TCR}\beta$ and from 3.88 to 1.37 on $\text{TCR}\alpha$ (Fig. 7). The residual signal was therefore publicity, not epitope-specific convergence.

5 Neighbourhood pgen (generation degree)

Point P_{gen} controls how likely a single CDR3 is to be generated, but chance homology depends on how many likely *neighbours* a sequence has. Following mirpy [6], we quantify this with the *1-mm neighbourhood pgen*—the generation probability of all CDR3s within Hamming distance 1, computed exactly with OLGA’s wildcard identity (`compute_hamming_dist_1_pgen`; terminal anchor positions skipped). Figure 8 shows the per-chain distributions.

The pgen-matched control also matches degree. immrep25 and its per-epitope P_{gen} -matched OLGA control (pgen-ladder) have essentially identical neighbourhood-pgen distributions (median $\Delta \log_{10} \approx 0.04$ on $\text{TCR}\alpha$, 0.07 on $\text{TCR}\beta$; orange and red curves overlap). Matching the point pgen therefore already matches the generation degree, so the control is valid for homology matchability and immrep25’s residual homology is *not* a neighbourhood-density artefact—consistent with its being publicity (§4), which is removed by chain, not by degree.

Degree spans a decade, yet homology S/N stays at the floor. Absolute degree differs strongly with how the real repertoire is sampled, and both extremes are shown. Sampled uniformly over *unique* clonotypes (AIRR random), the repertoire sits at *lower* neighbourhood pgen than

epitope	n	β nbrs	α nbrs	β S/N
YLFNADIWI	50	20	1	29.35
MTDYDYLEV	50	13	3	27.00
TVYPYGTSL	50	5	2	11.00
ILHTHVPEV	50	5	7	9.11
YMFYDGYDV	50	3	2	6.28
SQFNWTIYL	50	2	1	5.00
REDDYSVWL	50	2	1	4.52
SEESAFYVL	50	1	2	3.00
FLDCKSYIL	50	1	4	2.86
GEVLACYAL	50	1	1	2.83
FEAQPGALL	50	1	3	2.59
MEMPDYLLL	50	0	3	1.00
HLDDYPYLM	50	0	1	1.00
KLYPFLWFA	50	0	11	1.00
GENALTYAL	50	0	3	1.00
YENGSTPVL	50	0	1	0.95
YEDGVIFYL	50	0	5	0.91
ILMHATYFL	50	0	1	0.90
MENWSALEL	50	0	0	0.88
YESYIPGAL	50	0	1	0.83
immrep25 (geom. mean)	1000	54	53	2.69
AIRR random	1000	0	0	1.00
AIRR rank-ladder	1000	5	93	0.86
OLGA pgen-ladder	1000	0	1	0.95

Table 1: immrep25 positives by epitope (n TCRs) and their count of within-epitope Hamming-1 CDR3 neighbours per chain and β S/N, versus the structured/random controls. Sorted by β S/N.

pgen-weighted OLGA generation—its diversity is dominated by low-pgen singletons. Sampled as the top clonotypes by *clone frequency* (AIRR rank-ladder), the public, expanded clones give the *highest* neighbourhood pgen of any cohort (median ~ 0.9 dex above OLGA on TCR α), confirming that abundant real TCRs are indeed high-degree. Crucially, *both* AIRR variants sit at homology S/N ≈ 1 (Fig. 1): the S/N is normalised against a random re-partition of each cohort, so it measures epitope *specificity*, not absolute degree. Generation degree and epitope-specific convergence are therefore distinct axes—immrep25 is elevated on neither beyond publicity.

6 Conclusion

Across both probes and both chains the calibration cohorts recover the expected order, and the immrep25 positives fall at the bottom, just above the real-repertoire (AIRR) and OLGA floors and one to three orders of magnitude below every quality cohort (Fig. 9). Their weak homology is contributed by public TCRs and collapses to the floor when those are removed; their inter-chain pairing shows essentially no coupling. We therefore find *no meaningful epitope-specific signal—in homology or chain pairing—among the immrep25 positives beyond noise and publicity*, consistent with the benchmark’s own finding that models do not exceed random guessing on unseen peptides [3].

Relation to reported prediction scores. The earlier IMMREP23 challenge [7] was built largely on curated VDJdb-quality data and reported that specificity can be predicted for its *seen* peptides with useful accuracy (median $\text{AUC}_{0.1} \gtrsim 0.7$)—consistent with our finding that the VDJdb cohorts carry strong homology and pairing signal (Figs. 1, 5). IMMREP25, in contrast, found no method exceeded a best macro- $\text{AUC}_{0.1}$ of 0.60 on *unseen* peptides [3]. Our audit

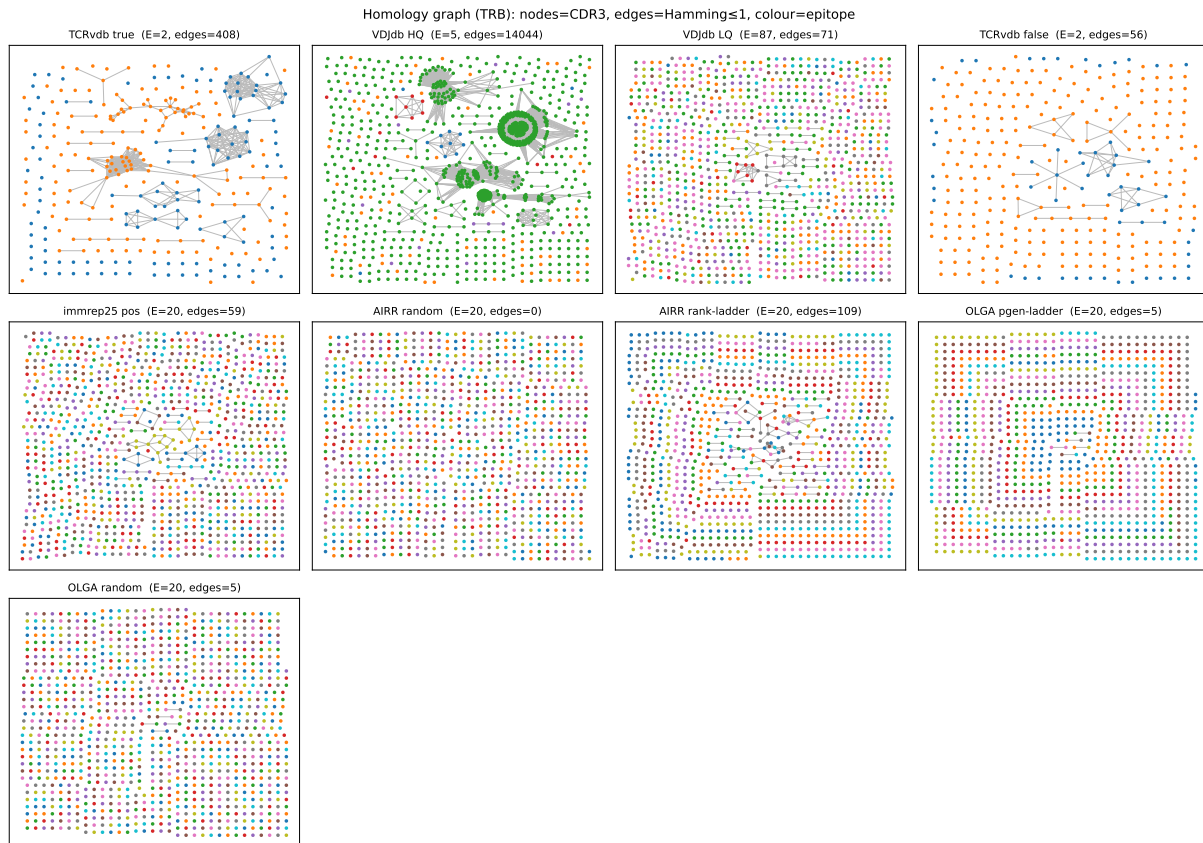


Figure 4: TCR β homology graph per cohort (all epitopes ≥ 30 records). Verified cohorts form dense epitope-coloured cliques; the controls show no epitope-coloured clustering (the AIRR rank-ladder has edges from public clones, but they cross colours).

accounts for the gap without running any predictor: the unseen-peptide positives carry no epitope-specific signal beyond publicity, so a model has nothing epitope-specific to latch onto and can only exploit similarity to previously seen, VDJdb-like TCRs—which novel epitopes, by construction, do not provide.

Caveats. TCRvdb true/false separate less than expected because p_{adj} scores differential enrichment in one screen rather than absence of specificity, so “false” still contains epitope-associated TCRs; and TCRvdb and VDJdb-HQ rest on only 2 and 5 epitopes respectively, so their absolute S/N is uncertain (wide CIs). Neither affects immrep25, whose 20 epitopes place it just above the real-repertoire (AIRR) and OLGA floors on every probe. Reproduce with the fetch scripts (`fetch_vdjdb.sh`, `fetch_airr.sh`, `gen_olga_pool.sh`), then `build_olga.py`, `build_airr.py`, `run_audit.py`, `make`.

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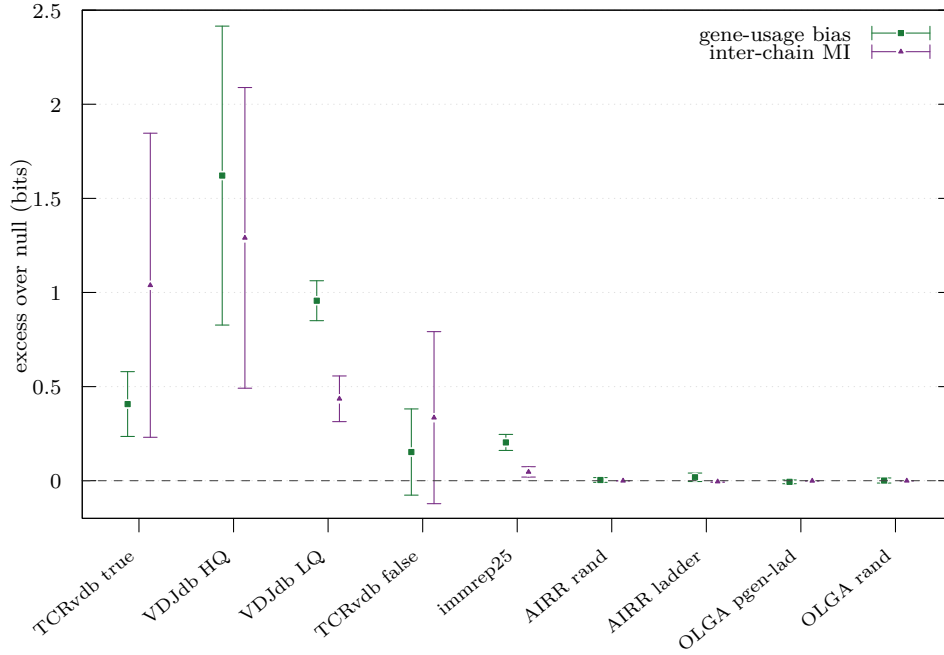


Figure 5: Pairing signals per cohort (one-vs-many over all epitopes with ≥ 30 records): per-epitope excess over the permutation null, in bits (mean $\pm$ 95% CI). Squares = gene-usage bias, triangles = inter-chain MI. Dashed line = 0 (random). immrep25 and all controls sit at the random line, especially for MI.

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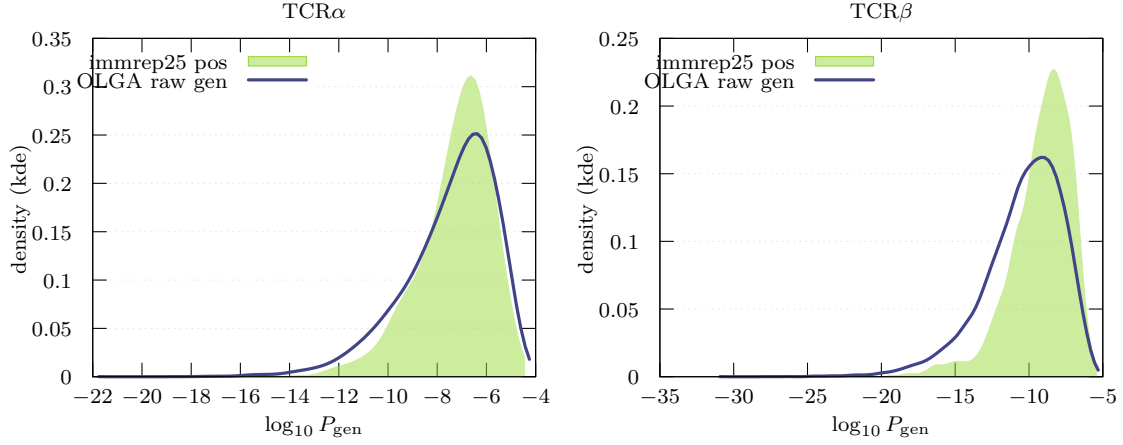


Figure 6: Per-chain $\log_{10} P_{\text{gen}}$ (kernel density): immrep25 positives (filled) vs. raw OLGA generation (line). Positives skew to higher P_{gen} (public-leaning), motivating the P_{gen} -matched control.

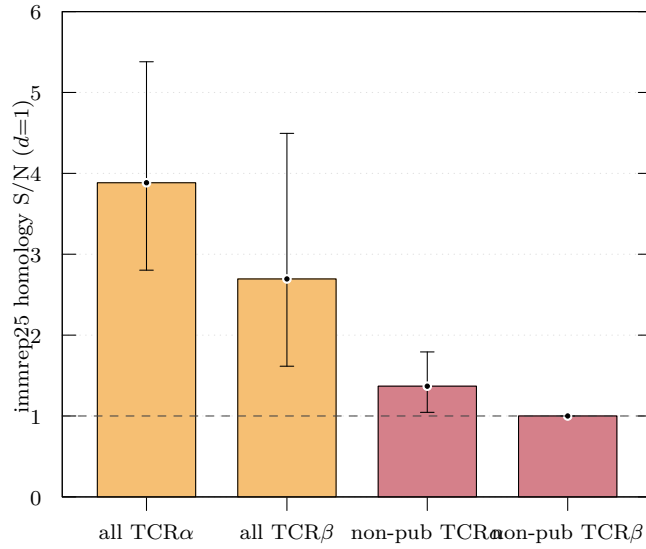


Figure 7: immrep25 homology S/N ($d=1$) for all positives vs. non-public positives (Hamming ≤ 1 to references removed). Removing public TCRs drives S/N to the noise floor (dashed line).

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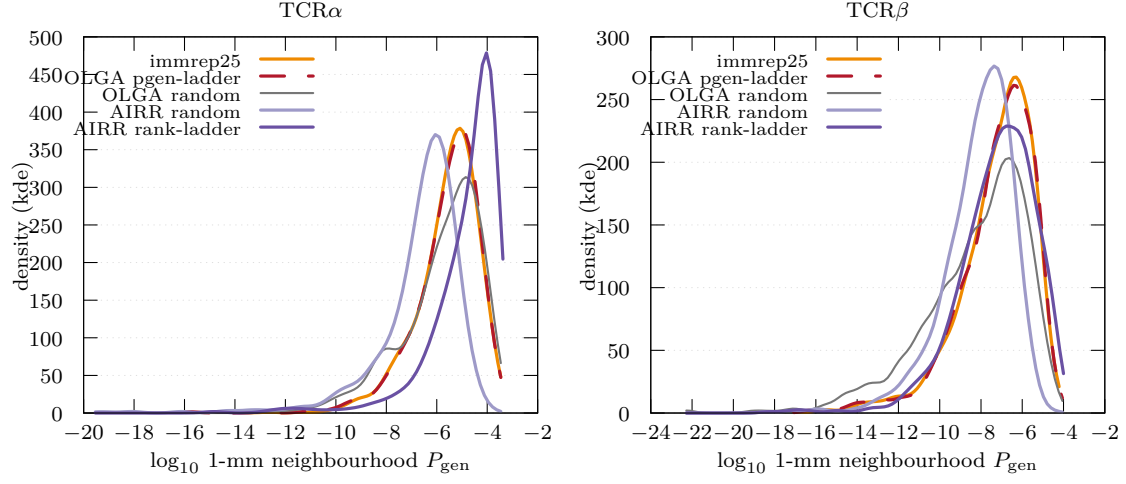


Figure 8: 1-mm neighbourhood P_{gen} (kernel density) per chain. immrep25 overlaps its pgen-matched OLGA control (degree is controlled). The real AIRR repertoire spans a decade of degree by sampling—lowest for unique clonotypes, highest for the abundant top-frequency clones—yet both sit at homology $S/N \approx 1$ (Fig. 1).

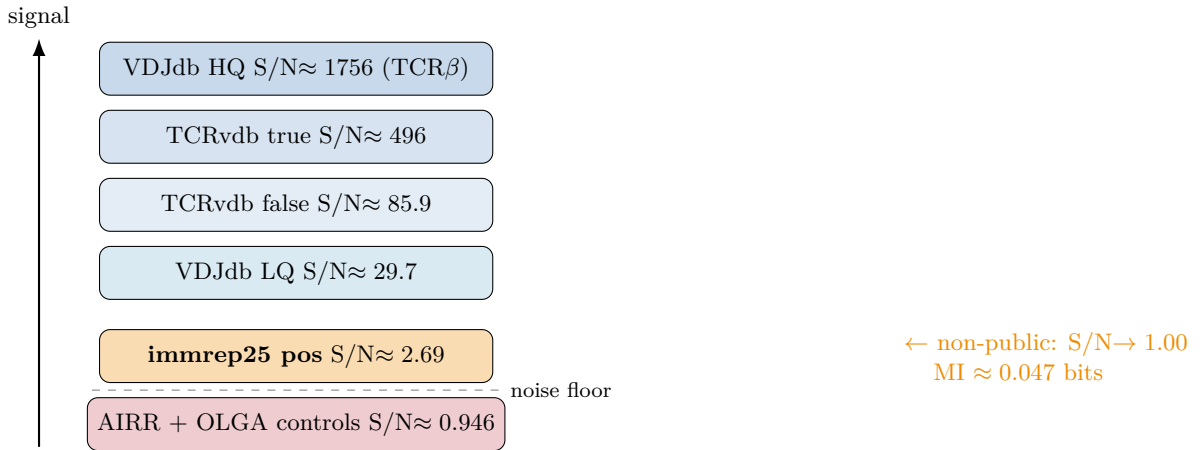


Figure 9: Signal-to-noise ladder (TCR β homology, one-vs-many). The immrep25 positives sit one rung above the random floor; stripping public TCRs lowers them onto it.